

Pull soft from hotspots

Net weighting raises the cost of nets that cross congested GCells so a weighted placer pulls those pins apart

The idea

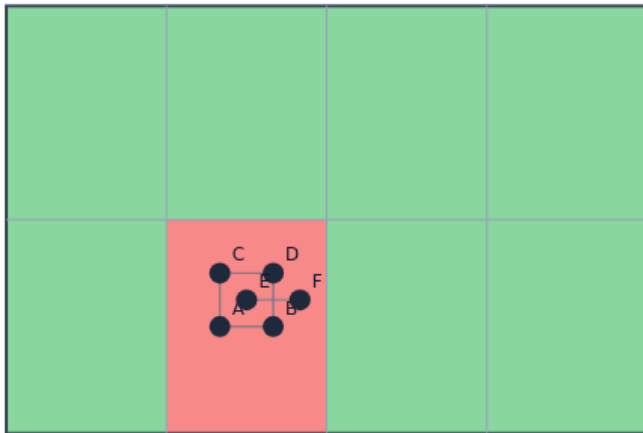
- For each net, average congestion over GCells under its bbox
- Scale with beta, one point zero is a clear demo
- The four-pin net on a clustered seed should outrank the short E-F net

Weight hot nets

NET WEIGHTING

Weight hot nets

Step 1 / 5 · idea · module03-03-net-weighting



Explanation

Nets through congested GCells get larger weights for weighted place.

- $w=1+\beta \cdot \text{mean cong}$

$\beta=1$ demo

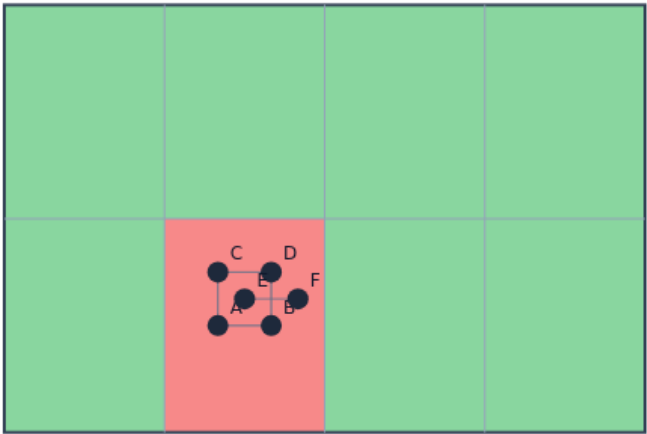
Weight hot nets

Mean under bbox

NET WEIGHTING

Mean under bbox

Step 2 / 5 · bbox · module03-03-net-weighting



Explanation

Average congestion over GCells under the net bbox—not the whole chip.

- Local mean

6 nets

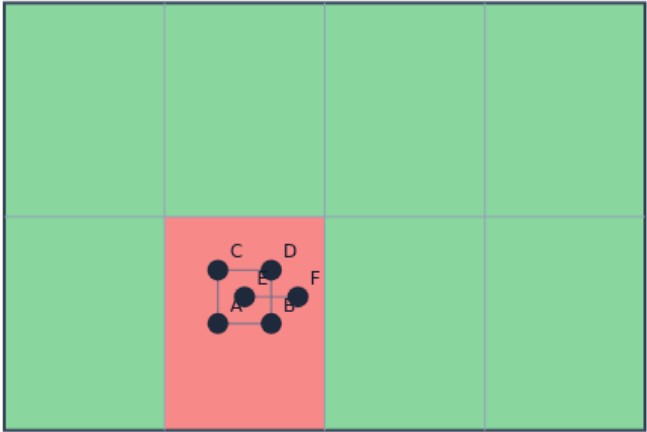
Mean under bbox

4-pin ranks high

NET WEIGHTING

4-pin ranks high

Step 3 / 5 · rank · module03-03-net-weighting



Explanation

On a cluster, the 4-pin net outranks short E-F.

- $w_4 \geq w_5$

Timing cousin

4-pin ranks high

Spread lowers weights

NET WEIGHTING

Spread lowers weights

Step 4 / 5 · cool · module03-03-net-weighting

The diagram shows a 2x4 grid. The first three columns are red, and the fourth is green. Nodes A, B, C, and D are at the corners of the first three columns, connected by lines. Nodes E and F are in the middle of the second and third columns. A light gray box contains the text 'Still report overflow'.

Explanation

As congestion falls, weights ease back toward 1.

- Coupled to map

Still report overflow

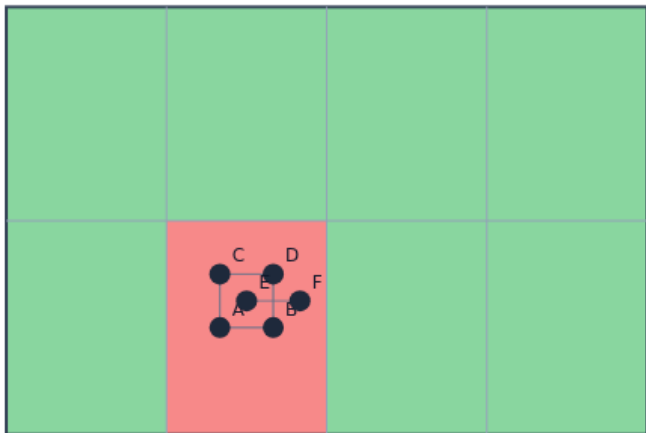
Spread lowers weights

Use in placer

NET WEIGHTING

Use in placer

Step 5 / 5 · use · module03-03-net-weighting



Explanation

Weighted HPWL pulls soft from hotspots—pair with inflators.

- Two knobs

Feedback lab next

Use in placer

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Congestion-aware net weighting

INTERACTIVE TOOL

Congestion-aware net weighting

Raise weights for nets whose bbox covers congested GCells.

Your job: weights = 1 + mean congestion under each net bbox. Keep the 4-pin net heavier than E-F on a hot seed.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntroSix weights

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)
HPWL: 7.0
overflow total: 5.00 · max: 5.00 · count: 1
capacity: 2
view: your placement
weights: 4.50. 4.50. 4.50. 4.50. 4.50. 4.50

Browser lab starter

learn_congestion — 03 net weighting

Implement track

- Implement ``net_weights_from_congestion``
- Print the six weights for `congested_seed`
- Confirm net index four is among the highest

Pitfalls

- Averaging over the whole chip instead of the net bbox
- Using demand instead of congestion ratios
- Updating weights but forgetting the placer still optimizes unweighted HPWL in the toy lab

Your turn

- Finish weighting
- Next: one full placement feedback pass